

Project Title:- Sales Data Analysis Project

Project Overview

Dataset Information

Tools and Techniques Used

Project Phases

Business Questions and Answers

Dashboard Screenshots

Key Insights and Recommendations

Challenges and Solutions

Limitations

Future Scope

Conclusion

Overview:

This project analyzes retail sales data to uncover revenue trends and business insights. The dataset was sourced from Kaggle and contains approximately 185,000 rows of sales transaction records covering products, cities, order dates, and revenue.

The goal of this project was to clean the raw data, engineer useful features, analyze business performance across months, cities, and products, and present findings through an interactive Excel dashboard.

Dataset Information

Source: Kaggle (Public Dataset)

Total Rows: 185,000 (approximately)

Original Columns (11):

Order ID, Product, Quantity Ordered, Price Each, Sales, Order Date, Purchase Address, City, Hour, Month, Year

Helper Columns Added (5):

Date Only, Year Only, Month Only, Quarter Only, Day Only

Time Period: January 2019 — December 2020 (2 years of data)

Purpose of Helper Columns: The Order Date column contained full datetime values. To enable monthly, quarterly, and day-wise analysis, five helper columns were extracted from this column using Excel functions — YEAR(), MONTH(), TEXT(), and ROUNDUP().

Tools and Techniques Used

Primary Tool: Microsoft Excel

Data Cleaning:

- Conditional Formatting — to highlight duplicate records visually
- COUNTIFS formula — to create a Duplicate_Check column and identify true duplicates to delete.
- Excel Filter — to isolate and delete duplicate rows

Feature Engineering:

- INT(F2) — to extract Date Only from Order Date
- YEAR(F2) — to extract Year
- TEXT(F2,"mmmm") — to extract Month Name, if we use Month(f2)→ 1, 2, 3... 12 useful as a number but not as a name .
- TEXT(F2,"dddd") — to extract Day of Week
- "Q"&ROUNDUP(MONTH(F2)/3,0) — to extract Quarter

Analysis:

- **Pivot Tables** — for Total Revenue, Total Orders, Top City, Top Month, Monthly Revenue, Highest City Revenue, Product Revenue

Visualization:

- **Pivot Charts** — Bar charts for city, product, and monthly revenue trends
- **Excel Dashboard** — dedicated sheet combining all KPI cards and charts

Project Phases--

Phase 1: Dataset Understanding:

Imported the raw dataset containing 185,000 approximate rows and 11 columns. Explored all columns, checked for missing values, identified duplicate records, and created a Data Dictionary sheet to document column names, data types, and observations.

Phase 2: Data Cleaning

- Used Conditional Formatting to visually highlight duplicate records.
- Created a Duplicate_Check column using the COUNTIFS formula to programmatically identify duplicates.
- Filtered and deleted duplicate rows, retaining only unique records in a cleaned dataset.
- Verified Sales calculations
- Created a clean version of the dataset (Sales = Quantity × Price verified)

Phase 3: Feature Engineering

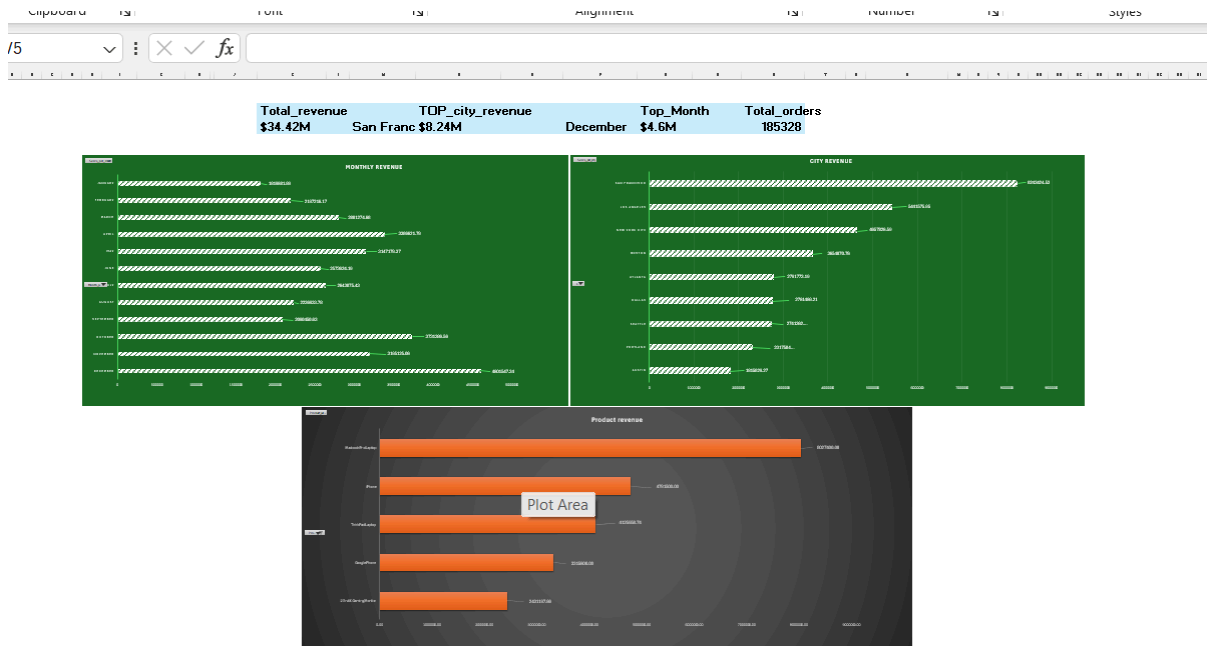
Extracted five new helper columns from the Order Date column — Date Only, Year Only, Month Only, Quarter Only, and Day Only — using Excel functions INT(), YEAR(), TEXT(), and ROUNDUP(). These columns enabled time-based analysis that was not possible with the original datetime format.

Phase 4: Exploratory Data Analysis

Built Pivot Tables to answer six business questions covering monthly revenue trends, city-wise performance, product revenue, units sold, peak order hours, and day-wise revenue. Each question was documented with its result and business insight.

Phase 5: Dashboard Creation

Designed an Excel dashboard combining four KPI cards (Total Revenue, Total Orders, Top City, Top Month) and three bar charts (Monthly Revenue, City Revenue, Product Revenue) to present findings in a single, readable view.



Business Questions and Answers

1: Which Month Generated the Highest Revenue?

Method Used:

Created a Pivot Table.
 Rows = Month_only
 Values = Sum of Sales

Result:

December generated the highest revenue with total sales of 4,601,547.34 (\$4.6M)

Business Insight:

December was the highest revenue-generating month. This may be due to **holiday** season shopping, year-end promotions, and increased consumer spending during the festive period.

D	F	G	H
	Row Labels	Sum of Sales	
	December	4601547.34	
	October	3731386.56	
	April	3386621.79	
	November	3195125.06	
	May	3147179.37	
	March	2801274.68	
	July	2643875.43	
	June	2573624.19	
	August	2236933.78	
	February	2197216.17	
	September	2090450.63	
	January	1819681.89	
	Grand Total	34424916.89	

PivotTable Fields

Choose fields to add to report:

Search

☐ SN
☐ Order ID
☐ Product
☐ Quantity Ordered
☐ Price Each
☐ Order Date
☐ Date_Only
☐ Year_only
☐ Quarter_only
☐ Day_only
☐ Purchase Address
☐ Month
☒ Month_only
☒ Sales
☐ City
☐ Hour
More Tables...

Drag fields between areas below:

Filters

Columns

Rows
Month_only

Σ Values
Sum of Sales

Question 2: Which City Generated the Highest Revenue?

Method Used:

Used Pivot Table.

Rows = City

Values = Sum of Sales

Result:

1. San Francisco 8,243,424.53
2. Los Angeles 5,441,575.95
3. New York City 4,657,826.59
4. Boston 3655508.26
5. Atlanta 2792843.19

Business Insight:

San Francisco generated the highest revenue among all cities. This indicates strong customer demand and purchasing activity in that market compared to other cities.

Row Labels	Sum of Sales
San Francisco	8243424.53
Los Angeles	5441575.95
New York City	4657826.59
Boston	3654070.76
Atlanta	2791773.19
Dallas	2761469.21
Seattle	2741362.65
Portland	2317584.74
Austin	1815829.27
Grand Total	34424916.89

Search

☐ SN
☐ Order ID
☐ Product
☐ Quantity Ordered
☐ Price Each
☐ Order Date
☐ Date_Only
☐ Year_only
☐ Quarter_only
☐ Day_only
☐ Purchase Address
☐ Month
☐ Month_only
☒ Sales
☒ City
☐ Hour
More Tables...

Filters

Columns

Rows
City

Σ Values
Sum of Sales

Question 3: Which Product Generated the Highest Revenue?

Method Used:

Created a Pivot Table.

Rows = Product

Values = Sum of Sales

Sorted values from Largest to Smallest.

Result:

1. Macbook Pro Laptop = 8,027,400
2. iPhone = 4,791,500
3. ThinkPad Laptop = 4,125,958.74

Business Insight:

High-priced products generated the highest revenue even though they were sold in lower quantities. Product price has a significant impact on revenue generation.

Row Labels	Sum of Sales
Macbook Pro Laptop	8027400
iPhone	4791500
ThinkPad Laptop	4125958.74
Google Phone	3315600
27in 4K Gaming Moni	2431197.66
34in Ultrawide Monitc	2350238.15
Apple AirPods Headpl	2341950
Flatscreen TV	1442100
Bose SoundSport Hea	1339566.03
27in FHD Monitor	1129724.68
Vareebadd Phone	827200
20in Monitor	453488.77
LG Washing Machine	399600
LG Dryer	387600
Lightning Charging Ca	341039.4
USB-C Charging Cabl	279582.2
Wired Headphones	245615.15
AA Batteries (4-pack)	105757.44
AAA Batteries (4-pack)	89798.67
Grand Total	34424916.89

Choose fields to add to report:

Search

☐ SN
☐ Order ID
☒ Product
☐ Quantity Ordered
☐ Price Each
☐ Order Date
☐ Date_Only
☐ Year_only
☐ Quarter_only
☐ Day_only
☐ Purchase Address
☐ Month
☐ Month_only
☒ Sales
☐ City
☐ Hour
More Tables...

Drag fields between areas below:

Filters

Columns

Rows
Product

Σ Values
Sum of Sales

Question 4: Which Product Sold the Most Units?

Method Used:

Pivot Table.

Rows = Product

Values = Sum of Quantity Ordered

Sorted values from Largest to Smallest.

Result:

1. AAA Batteries (4-pack) = 30,880 units

2. AA Batteries (4-pack) = 27,541 units
3. USB-C Charging Cable = 23,873 units

Business Insight:

Low-cost products and accessories were purchased more frequently than expensive products. Customers regularly buy batteries and charging accessories, resulting in higher sales volume.

Row Labels	Sum of Quantity Ordered
AAA Batteries (4-pack)	30880
AA Batteries (4-pack)	27541
USB-C Charging Cable	23873
Lightning Charging Cable	23103
Wired Headphones	20485
Apple Airpods Headphon	15613
Bose SoundSport Headpl	13403
27in FHD Monitor	7532
iPhone	6845
27in 4K Gaming Monitor	6234
34in Ultrawide Monitor	6185
Google Phone	5526
Flatscreen TV	4807
Macbook Pro Laptop	4722
ThinkPad Laptop	4126
20in Monitor	4123
Vareebadd Phone	2068
LG Washing Machine	666
LG Dryer	646
Grand Total	208378

add to report:

Search

☐ SN
☐ Order ID
☒ Product
☒ Quantity Ordered
☐ Price Each
☐ Order Date
☐ Date_Only
☐ Year_only
☐ Quarter_only
☐ Day_only
☐ Purchase Address
☐ Month
☐ Month_only
☐ Sales
☐ City
☐ Hour
More Tables...

below:

Filters

Columns

Rows

Product

Σ Values

Sum of Quantity Orde...

Question 5: At What Hour Do Customers Place the Most Orders?

Method Used:

Created a Pivot Table.

Rows = Hour

Values = Count of Order ID

Sorted values from Largest to Smallest.

Result:

1. Hour 19 (7 PM) = 12,859 orders
2. Hour 12 (12 PM) = 12,551 orders

3. Hour 11 (11 AM) = 12,369 orders

Business Insight:

Customer purchasing activity peaks during lunch hours and evening hours.

Businesses can schedule advertisements, promotions, and notifications during these periods to increase customer engagement and sales.

D	F	G	H
	Row Labels	Count of Order ID	
	19	12859	
	12	12551	
	11	12369	
	18	12240	
	20	12208	
	13	12095	
	14	10942	
	10	10904	
	21	10881	
	17	10865	
	16	10332	
	15	10133	
	22	8792	
	9	8726	
	8	6248	
	23	6245	
	7	3989	
	0	3894	
	6	2476	
	1	2344	
	5	1317	
	4	1244	

PivotTable Fields

Choose fields to add to report:

Search

- ☐ SN
- ☒ Order ID
- ☐ Product
- ☐ Quantity Ordered
- ☐ Price Each
- ☐ Order Date
- ☐ Date_Only
- ☐ Year_only
- ☐ Quarter_only
- ☐ Day_only
- ☐ Purchase Address
- ☐ Month
- ☐ Month_only
- ☐ Sales
- ☐ City
- ☒ Hour

More Tables...

Drag fields between areas below:

Filters

Columns

Rows

Hour

Values

Count of Order ID

Question 6: Which Day Generated the Highest Revenue?

Method Used:

Created a Pivot Table.

Rows = Day_only

Values = Sum of Sales

Sorted values from Largest to Smallest.

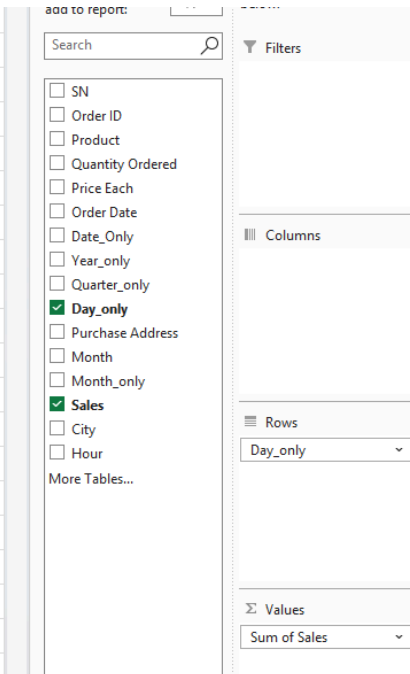
Result:

1. Tuesday = 5,082,518.01
2. Wednesday = 4,982,646.28
3. Sunday = 4,920,381.93

Business Insight:

Revenue is relatively high during the early and middle part of the week. Tuesday generated the highest revenue, while Sunday also showed strong performance, indicating consistent customer demand throughout the week.

Row Labels	Sum of Sales
Tuesday	5082518.01
Wednesday	4982646.28
Sunday	4920381.93
Saturday	4894161.61
Monday	4869793.38
Friday	4849223.99
Thursday	4826191.69
Grand Total	34424916.89



The image shows a screenshot of a Power BI report. On the left is a table with two columns: 'Row Labels' and 'Sum of Sales'. The table lists the days of the week from Tuesday to Thursday, followed by a 'Grand Total' row. The 'Sum of Sales' values are: Tuesday (5082518.01), Wednesday (4982646.28), Sunday (4920381.93), Saturday (4894161.61), Monday (4869793.38), Friday (4849223.99), and Thursday (4826191.69). The 'Grand Total' is 34424916.89. On the right is a sidebar with a search bar, a list of fields (SN, Order ID, Product, Quantity Ordered, Price Each, Order Date, Date_Only, Year_only, Quarter_only, Day_only, Purchase Address, Month, Month_only, City, Hour), and sections for 'Columns', 'Rows', and 'Values'. The 'Day_only' field is selected in the 'Rows' section, and 'Sum of Sales' is selected in the 'Values' section.

KPI cards for dashboard:

If a manager opens my dashboard for 10 seconds, what are the 4 most important numbers they should see immediately?

KPI cards: Quick Summary metrics of business performance

Total Revenue (\$34.42M)

Top Revenue City (San Francisco - \$8.24M)

Best Revenue Month (December - \$4.6M)

Total Orders (185,328)

Analytical Charts(Middle section): These help to compare performance and identify trends.

- Monthly Revenue Analysis

- City Revenue Analysis

Purpose:

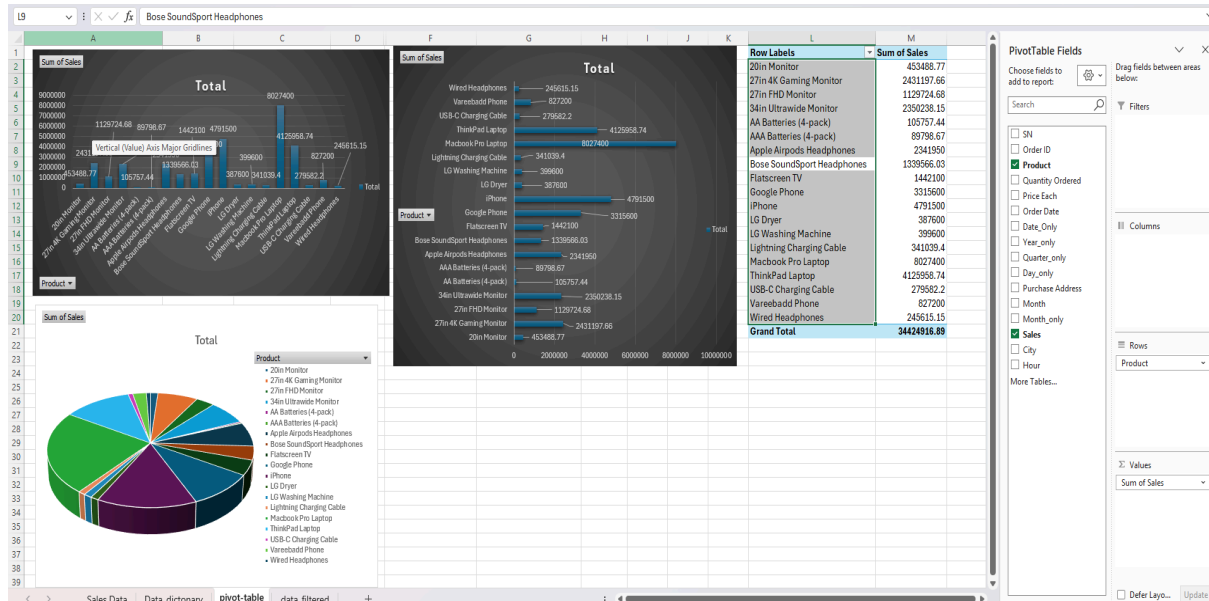
- To Identify seasonal sales trends
- To Compare revenue across cities
- Highlight top-performing locations and months

Product Performance Chart:

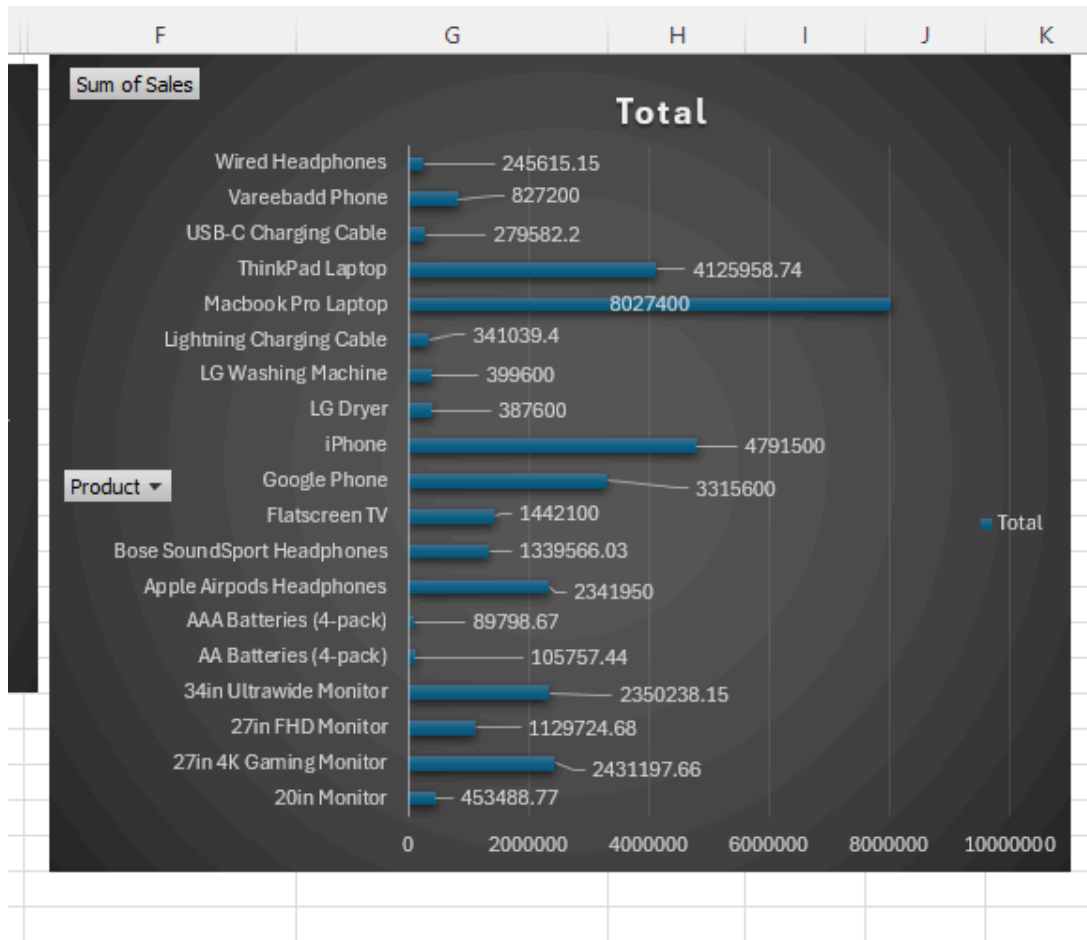
Top 5 Products by Revenue

Chart selection:

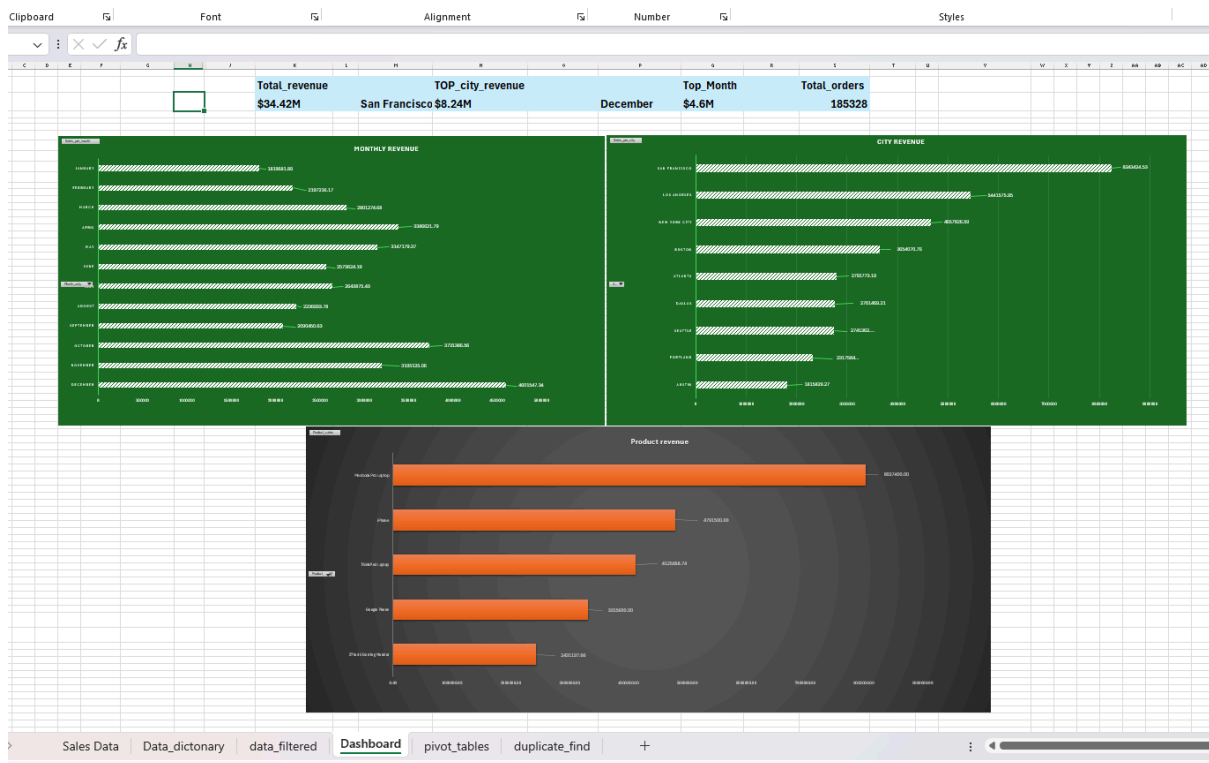
I tested all three charts(column, bar, pi chart).
I compared them.
I found Bar Chart easier to understand.



Bar chart



Final Dashboard:



Key Insights and Recommendations

Insight 1 — Seasonal Revenue Peak

December generated the highest monthly revenue of \$4.6M, driven by holiday season shopping and festive demand. The business should increase product inventory in November and launch targeted promotional campaigns before December to capture early holiday shoppers and maximize peak season revenue.

Insight 2 — Geographic Market Strength

San Francisco generated the highest city revenue at \$8.24M, significantly ahead of other cities. The business should prioritize marketing spend, inventory allocation, and customer retention efforts in San Francisco while investigating growth opportunities in underperforming cities like Atlanta and Austin.

Insight 3 — Price vs Volume Trade-off

High-priced products like MacBook Pro Laptop (\$8.02M revenue) generated the most revenue despite selling fewer units, while low-cost products like AAA Batteries sold the most units (30,880) but generated minimal revenue. The business should maintain a balanced product strategy — premium products drive revenue while accessories drive order volume and customer engagement.

Insight 4 — Peak Purchase Hours

Customer order activity peaks at 12PM (lunch break) and 7PM (post work hours). The business should schedule promotional notifications, discount offers, email campaigns, and advertisements to be delivered during these windows to maximize customer engagement and conversion rates.

Challenges and Solutions

Challenge 1 — Identifying True Duplicates

Initial duplicate detection using Conditional Formatting revealed that some Order IDs appeared multiple times, but on closer inspection, not all repeated Order IDs were true duplicates — some had different products under the same Order ID (representing a multi-item order), while others had identical Order ID, Product, Quantity, Price, and Address (true duplicate entries).

Solution: Created a Duplicate_Check column using the COUNTIFS formula(=COUNTIFS(B:B,B873,C:C,C873)) to flag rows where Order ID, Product, Price, and Address all matched exactly. Filtered the dataset to isolate rows marked as duplicates (value = 2), deleted them, then re-filtered to confirm only unique records remained (value = 1). This ensured legitimate multi-item orders were preserved while true duplicate entries were removed.

SN	Order ID	Product	Duplicate check	Quantity	Order Price	Order Date	Purchase Address	Sales	City	Region
873	874	AA Batteries (4-pack)	1	1	3.84	17-01-2019 23:02	131 2nd St, Boston, MA 02215	1	3.84	Boston
874	875	AA Batteries (4-pack)	1	1	3.84	23-02 131 2nd St, Boston, MA 02215	1	3.84	Boston	23
4107	4125	Lightning Charging Cable	2	1	14.95	06-01-2019 03:01	182 Jefferson St, San Francisco, CA 94016	1	14.95	San Francisco
4108	4126	Lightning Charging Cable	2	1	600	21-01-2019 11:23	918 Highland St, New York City, NY 10001	1	600	New York City
5786	5810	Google Phone	2	1	11.99	04-01-2019 16:50	883 4th St, Dallas, TX 75001	1	11.99	Dallas
5787	5811	Google Phone	2	1	11.99	04-01-2019 16:50	883 4th St, Dallas, TX 75001	1	11.99	Dallas
6778	6806	Wired Headphones	2	1	11.95	08-01-2019 17:36	562 14th St, Boston, MA 02215	1	11.95	Boston
6779	6807	Wired Headphones	2	1	11.95	08-01-2019 17:36	562 14th St, Boston, MA 02215	1	11.95	Boston
8102	8133	USB-C Charging Cable	2	1	14.95	12-01-2019 12:30	180 1st St, Boston, MA 02215	1	14.95	Boston
8103	8134	USB-C Charging Cable	2	1	14.95	12-01-2019 12:30	180 1st St, Boston, MA 02215	1	14.95	Boston
8274	8308	Lightning Charging Cable	2	1	14.95	12-01-2019 12:30	180 1st St, Boston, MA 02215	1	14.95	Boston
8275	8309	Lightning Charging Cable	2	1	150	02-01-2019 23:07	351 Madison St, New York City, NY 10001	1	150	New York City
8435	8469	Apple AirPods Headphones	2	1	150	02-01-2019 23:07	351 Madison St, New York City, NY 10001	1	150	New York City
8436	8470	Apple AirPods Headphones	2	1	11.95	14-01-2019 21:19	913 10th St, Los Angeles, CA 90001	1	11.95	Los Angeles
8653	8689	USB-C Charging Cable	2	1	11.95	14-01-2019 21:19	913 10th St, Los Angeles, CA 90001	1	11.95	Los Angeles
8654	8690	USB-C Charging Cable	2	1	11.95	14-01-2019 21:19	913 10th St, Los Angeles, CA 90001	1	11.95	Los Angeles
8886	8922	USB-C Charging Cable	2	1	11.95	11-01-2019 11:22	612 West St, New York City, NY 10001	1	11.95	New York City
8887	8923	USB-C Charging Cable	2	1	11.95	11-01-2019 11:22	612 West St, New York City, NY 10001	1	11.95	New York City
9388	9426	Wired Headphones	2	1	11.99	21-01-2019 09:20	691 Pine St, San Francisco, CA 94016	1	11.99	San Francisco
9389	9427	Wired Headphones	2	1	11.99	21-01-2019 09:20	691 Pine St, San Francisco, CA 94016	1	11.99	San Francisco
10111	431	Lightning Charging Cable	2	1	14.95	06-02-2019 16:07	111 10th St, Austin, TX 73301	2	14.95	Austin
10112	432	Lightning Charging Cable	2	1	14.95	06-02-2019 16:07	111 10th St, Austin, TX 73301	2	14.95	Austin
10120	440	iPhone	2	1	700	07-02-2019 17:43	784 Elm St, Boston, MA 02215	2	700	Boston
10122	442	iPhone	2	1	700	07-02-2019 17:43	784 Elm St, Boston, MA 02215	2	700	Boston
10140	460	USB-C Charging Cable	2	1	11.95	06-02-2019 19:13	759 1st St, Austin, TX 73301	2	11.95	Austin
10141	461	USB-C Charging Cable	2	1	11.95	06-02-2019 19:13	759 1st St, Austin, TX 73301	2	11.95	Austin
10227	547	Wired Headphones	2	1	11.99	19-02-2019 08:38	35 Pine St, Portland, OR 97035	2	11.99	Portland
10228	548	Wired Headphones	2	1	11.99	19-02-2019 08:38	35 Pine St, Portland, OR 97035	2	11.99	Portland
10843	1163	USB-C Charging Cable	2	1	11.95	25-02-2019 19:29	666 Meadow St, Boston, MA 02215	2	11.95	Boston
10844	1164	USB-C Charging Cable	2	1	11.95	25-02-2019 19:29	666 Meadow St, Boston, MA 02215	2	11.95	Boston
10903	1223	Wired Headphones	2	1	11.99	10-02-2019 21:52	504 Center St, Dallas, TX 75001	2	11.99	Dallas
10904	1224	Wired Headphones	2	1	11.99	10-02-2019 21:52	504 Center St, Dallas, TX 75001	2	11.99	Dallas
11096	1416	USB-C Charging Cable	2	1	11.95	06-02-2019 12:11	475 Jackson St, San Francisco, CA 94016	2	11.95	San Francisco
11097	1417	USB-C Charging Cable	2	1	11.95	06-02-2019 12:11	475 Jackson St, San Francisco, CA 94016	2	11.95	San Francisco
11592	1917	Bose SoundSport Headphone	2	1	99.99	25-02-2019 18:53	827 Dogwood St, Los Angeles, CA 90001	2	99.99	Los Angeles
11593	1918	Bose SoundSport Headphone	2	1	99.99	25-02-2019 18:53	827 Dogwood St, Los Angeles, CA 90001	2	99.99	Los Angeles
12604	2936	Wired Headphones	2	1	11.99	12-02-2019 20:08	74 Meadow St, Austin, TX 73301	2	11.99	Austin
12605	2937	Wired Headphones	2	1	11.99	12-02-2019 20:08	74 Meadow St, Austin, TX 73301	2	11.99	Austin

Learning: Working with a 185,000-row dataset, filtering and sorting operations took significant processing time. This taught the importance of verifying data carefully before deletion, since duplicate removal is irreversible.

Challenge 2 — Choosing the Right Chart Type

With no prior experience selecting chart types for business dashboards, initial attempts used pie charts and column charts to represent product, city, and monthly revenue data.

Solution: Tested the same dataset across pie, column, and bar charts side by side. Pie charts became unreadable with many categories, and column charts caused label overlap when revenue values were in the millions. Bar charts proved most effective — category names remained fully readable, and value labels did not overlap.

Learning: Chart selection should be driven by data characteristics (number of categories, value size, label length) rather than personal preference.

Limitations

This analysis is based on a dataset covering only 2019–2020 (2 years), so longer-term trends across multiple years could not be evaluated.

Additionally, the dataset includes only a limited set of cities, meaning the geographic insights reflect performance within these specific markets only and cannot be generalized to represent nationwide sales patterns. Expanding the dataset to include more cities and a longer time range would allow for more comprehensive trend analysis and broader business recommendations.

Future Scope

The next phase of this project involves recreating the same analysis using SQL to demonstrate the same business questions can be answered through database queries, and to compare the efficiency of Excel versus SQL approaches.

Conclusion

This project analyzed retail sales data through a complete workflow — from raw data exploration and cleaning, to feature engineering, business analysis, and dashboard visualization.

Duplicate records were carefully identified and removed using a verification-based filtering approach rather than direct deletion, ensuring data accuracy.

Pivot tables were used to answer six key business questions covering revenue trends, city performance, product performance, and customer purchase behavior.

The findings were consolidated into an interactive Excel dashboard highlighting the most important KPIs and top 5 rankings, designed for quick, clear business decision-making.

Section 1 — SQL Project Overview:

This SQL project uses the same cleaned sales dataset from the Excel analysis phase, converted to CSV format for import into MySQL.

The goal is to answer the same six business questions using SQL queries instead of Excel Pivot Tables, and to demonstrate that data analysis can be performed effectively using both tools. The database `sales_project` and table `sales_data` were created in MySQL Workbench to store and query the data.

Section 2 — Database and Table Setup

Database Name: `sales_project`

Table Name: `sales_data`

SQL TABLE:

```
CREATE TABLE sales_data(  
  
    ID INT AUTO_INCREMENT PRIMARY KEY,  
  
    Order_ID INT,  
  
    Product VARCHAR(50),  
  
    Quantity_Ordered INT,  
  
    Price_Each DECIMAL(10,2),  
  
    Order_Date DATETIME,  
  
    Purchase_Address VARCHAR(100),  
  
    Sales DECIMAL(10,2),  
  
    City VARCHAR(50),
```

Hour INT

);

Data Type Decisions:

- **ID** AUTO_INCREMENT PRIMARY KEY: ensures every row has a unique identifier, since Order_ID can repeat (one order can have multiple products).
- **Order_ID** INT: contains only whole numbers.
- **Product** VARCHAR(50): stores product name text.
- **Quantity_Ordered** INT: whole number units only.
- **Price_Each** DECIMAL(10,2): currency value requiring decimal precision.
- **Order_Date** DATETIME: stores both date and time in one column.
- **Purchase_Address** VARCHAR(100): full address strings can be long.
- **Sales** DECIMAL(10,2): currency value — DECIMAL preferred over FLOAT for financial data to avoid rounding errors.
- **City** VARCHAR(50): city name text.
- **Hour** INT: stores hour of day as a whole number (0-23).

Section 3 — Data Import Process

Step 1 — Why LOAD DATA INFILE instead of GUI Wizard

The MySQL Workbench Table Data Import Wizard was attempted first but stayed at 0% progress for over 20 minutes. The wizard is known to be slow and unstable for large files because it inserts rows one by one. **LOAD DATA INFILE** was used instead as it bulk-loads all rows in one operation — 185,328 rows imported in approximately 9 seconds.

Step 2 — Handling Helper Columns

The CSV file contained 16 columns including 7 helper columns (SN, Date_Only, Year_only, Quarter_only, Day_only, Month, Month_only) that were not needed in SQL since SQL can extract these values directly from `Order_Date` using built-in functions. The `@dummy` placeholder variables were used in the column mapping list to tell SQL to read but discard these columns during import:

```
(@dummy1, Order_ID, Product, Quantity_Ordered, Price_Each,  
Order_Date,
```

```
@dummy2, @dummy3, @dummy4, @dummy5, Purchase_Address,  
@dummy6, @dummy7,
```

```
Sales, City, Hour)
```

Step 3 — secure_file_priv Error and Fix

Running `LOAD DATA INFILE` with the original file path (`C:/Users/Prudhvi/OneDrive/Documents/`) produced Error 1290: secure_file_priv restriction. MySQL only allows reading files from one designated folder for security reasons. The allowed path was identified using:

```
SHOW VARIABLES LIKE 'secure_file_priv';
```

Result: `C:\ProgramData\MySQL\MySQL Server 9.6\Uploads\`

The CSV file was copied into this folder and the file path in the query was updated accordingly.

```

23      (@dummy1, Order_ID, Product, Quantity_Ordered, Price_Each
24
25 •   SHOW VARIABLES LIKE 'secure_file_priv';
26

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Variable_name	Value
secure_file_priv	C:\ProgramData\MySQL\MySQL Server 9.6\Uploads\

Step 4 — Order_Date Format Conversion

MySQL's DATETIME type requires the format **YYYY-MM-DD HH:MM**. The original CSV had dates in **DD-MM-YYYY HH:MM** format, which caused Error 1292 (Incorrect datetime value). The Order_Date column was converted in Excel using:

=TEXT(F2, "YYYY-MM-DD HH:MM:SS")

Since helper columns (Date_Only, Month_only, Year_only, Quarter_only) referenced the original Order_Date column via formulas, they were first converted to static values using Paste Special → Values Only before deleting the old column, to avoid breaking linked formulas.

Step 5 — 424 Truncation Warnings

After successful import, 424 warnings appeared: *"Data truncated for column Sales."* Investigation revealed some Sales values in the CSV had 3 decimal places (e.g. 99.999), which MySQL rounded to 2 decimal places (100.00) to fit the **DECIMAL(10, 2)** column definition. Three verification queries confirmed no data was actually damaged:

Test 1: Any Sales value too large for DECIMAL(10,2)?

```
SELECT Order_ID, Quantity_Ordered, Price_Each, Sales
```

```
FROM sales_data
```

```
WHERE Sales > 999999999;
```

Result: 0 rows

Test 2: Any Sales value silently converted to zero?

```
SELECT Order_ID, Sales
```

```
FROM sales_data
```

```
WHERE Sales = 0;
```

Result: 0 rows

Test 3: Any Sales value not matching Quantity × Price?

```
SELECT Order_ID, Quantity_Ordered, Price_Each, Sales,
```

```
       (Quantity_Ordered * Price_Each) AS calculated_sales
```

```
FROM sales_data
```

```
WHERE Sales != (Quantity_Ordered * Price_Each);
```

Result: 0 rows

All three checks returned zero rows, confirming all 185,328 Sales values are mathematically accurate. The warnings were harmless decimal rounding only.

Section 4 — SQL Business Questions

Question 1 — Which month generated the highest revenue?

Concept: `Order_Date` is a full DATETIME value. To group by month, the `MONTHNAME()` function extracts just the month name directly from the date column, no helper column needed, unlike Excel.

Query:

```
SELECT MONTHNAME(Order_Date), SUM(Sales)
FROM sales_data
GROUP BY MONTHNAME(Order_Date)
ORDER BY SUM(Sales) DESC;
```

Line by line:

- `MONTHNAME(Order_Date)` extracts the month name (January, February...) from the full datetime value.
- `SUM(Sales)` adds up all Sales values within each month group.
- `GROUP BY MONTHNAME(Order_Date)` combines all rows belonging to the same month into one group.
- `ORDER BY SUM(Sales) DESC` sorts results with highest revenue first.

Result:

→ December \$4,603,020.84
→ October \$3,732,695.61
→ April \$3,387,724.74

Business Insight: December generated the highest revenue, driven by holiday season shopping and festive demand. Businesses should increase inventory and run targeted promotions before December to maximize peak season revenue.

Question 2 — Which city generated the highest revenue.

Concept: `City` is already a direct column in the table, so no date function or special extraction is needed. The query simply groups all

rows by city name, sums the Sales within each city group, and sorts from highest to lowest to identify the top performing market.

Query:

SELECT City, SUM(Sales) as City_sales

FROM sales_data

GROUP BY City

ORDER BY SUM(Sales) **DESC**

Limit 5;

Result:

City	City_Sales
San Francisco	\$ 8247060.32
Los Angeles	\$ 5443980.13
New York City	\$ 4659246.09
Boston	\$ 3655508.26
Atlanta	\$ 2792843.19

Line by Line:

SELECT City - retrieves the city name for each group

SUM(Sales) AS City_sales: adds up all Sales values for each city and labels the result column as **City_sales** instead of displaying it as **SUM(Sales)**

SUM(Sales) - Simply renames a column in the output for readability

FROM sales_data : specifies the table to query

GROUP BY City : combines all rows belonging to the same city into one group so SUM() calculates one total per city

ORDER BY SUM(Sales) DESC : sorts results from highest to lowest revenue

LIMIT 5 : Returns only the top 5 cities instead of all cities

Business Insight:

San Francisco generated the highest city revenue at \$8.24M, significantly ahead of other cities. This indicates strong customer demand and purchasing power in that market. The business should prioritize marketing spend and inventory allocation in San Francisco while exploring growth opportunities in lower-performing cities like Atlanta and Austin.

Question 3 — Which product generated the highest revenue?

Concept: **Product** is already a direct column in the table, so no date function or special extraction is needed. **LIMIT 5** was used instead of showing all 20 products to keep results focused on top performers — the same approach used in the Excel dashboard where only the top 5 products were displayed in the chart.

Query:

SELECT Product, SUM(Sales) as Product_sales

FROM sales_data

GROUP BY Product

ORDER BY sum(Sales) DESC

LIMIT 5;

Result:

Product	Product_Sales
Macbook Pro Laptop	8027400.00
iPhone	4791500.00
ThinkPad Laptop	4125958.74
Google Phone	3315600.00
27in 4K Gaming Monitor	2431197.66

Line by Line:

SELECT Product, retrieves the product name from each group.

SUM(Sales) : adds up all Sales values per product and labels the result column as **Product_sales** for readability

FROM sales_data : specifies the table to query

GROUP BY Product : combines all rows belonging to the same **Product** into one group so **SUM()** calculates one total per product

ORDER BY sum(Sales) DESC: sorts results from highest to lowest revenue

LIMIT 5 : Returns only the top 5 Products instead of all 20 Products.

Business Insight: MacBook Pro Laptop generated the highest product revenue at \$8.02M despite selling fewer units compared to lower-priced products. This demonstrates that product price has a significant impact on revenue generation, a single high-value product can outperform many lower-priced products in revenue even when sold in smaller quantities. The business should maintain premium product offerings as they are the primary drivers of total revenue.

Question 4 — Which product sold the most units?

```
SELECT Product, sum(Quantity_Ordered) as Total_units  
  
FROM sales_data  
  
GROUP BY Product  
  
ORDER BY SUM(Quantity_Ordered) DESC  
  
LIMIT 5;
```

Result:

Product	Total_units
AAA Batteries (4-pack)	30880
AA Batteries (4-pack)	27541
USB-C Charging Cable	23873
Lightning Charging Cable	23103
Wired Headphones	20485

Line by Line:

SELECT Product : retrieves the product name from each group.

sum(Quantity_Ordered) **as** Total_units : It produces the sum of quantity ordered based on the product and produces the total quantity.

FROM sales_data : specifies the table to query

GROUP BY Product : combines all rows belonging to the same Product into one group so SUM() calculates one total per quantity

ORDER BY SUM(Quantity_Ordered) DESC : sorts results from highest to lowest quantity.

LIMIT 5 : Returns only the top 5 Products instead of all Products.

Business Insight: Low-cost accessories like AAA Batteries and AA Batteries sold the highest number of units, significantly more than premium products like MacBook Pro Laptop.

This reveals a clear price-volume trade-off, high-priced products drive revenue while low-cost products drive volume. The business should ensure consistent stock availability for high-volume accessories throughout the year to meet steady customer demand.

Question 5 — At what hour do customers place the most orders?

SELECT Hour, Count(DISTINCT Order_ID) **as**
number_of_orders

FROM sales_data

GROUP BY Hour

ORDER BY COUNT(DISTINCT Order_ID) DESC

LIMIT 5;

Result:

Hour	number_of_orders
19	12354
12	12064
11	11861

20 11753

18 11743

Line by Line:

SELECT Hour : retrieves the Order hour from each group.

Count(DISTINCT Order_ID) **as** number_of_orders : It produces the Count of Orders based on the Unique order_id

FROM sales_data : specifies the table to query

GROUP BY Hour : combines all rows belonging to the same Hour into one group so count() calculates total units sold per hour.

ORDER BY COUNT(DISTINCT Order_ID) DESC : sorts Order count results from highest to lowest quantity.

Limit 5: Returns only the top 5 hours with the highest number of orders.

Business Insight: Customer purchasing activity peaks during lunch hours (12 PM) and evening hours (7 PM). Businesses can schedule promotional campaigns, notifications, and limited-time offers during these periods to maximize customer engagement and conversion rates.

Question 6 — Which day generated the highest revenue?.

Concept: Order_Date is a full DATETIME value. The DAYNAME() function extracts the day name (Monday, Tuesday...) directly from the date column, similar to how MONTHNAME() was used for Question 1.

This eliminates the need for a helper column like `Day_only` that was created in Excel

Query:

```
SELECT DAYNAME(Order_Date) as day_name, SUM(Sales)  
as total_sales
```

```
FROM sales_data
```

```
GROUP BY DAYNAME(Order_Date)
```

```
ORDER BY SUM(Sales) DESC
```

```
LIMIT 5;
```

Result:

day_name	total_sales
Tuesday	5084397.91
Wednesday	4984571.18
Sunday	4922178.44
Saturday	4895954.77
Monday	4871789.08

Line by line:

```
SELECT DAYNAME(Order_Date) as day_name : Extracts the  
weekday name (Monday, Tuesday, etc.) from the Order_Date  
column.
```

SUM(Sales) as total_sales : It retrieves the total sales per day group it in a particular day base.

FROM sales_data : specifies the table to query

GROUP BY DAYNAME(Order_Date) : Combines all rows belonging to the same dayname into one group so SUM() calculates total sales per day.

ORDER BY SUM(Sales) DESC : sorts Order SUM results from highest to lowest quantity.

LIMIT 5 : Returns only the top 5 revenue-generating days instead of all days.

Business Insight: Tuesday generated the highest day-wise revenue at \$5.08M, with Wednesday and Sunday close behind.

Revenue remains consistently strong throughout the week with no significant low-performing days, indicating steady customer demand across all days. The business should ensure sufficient inventory and timely delivery capacity especially on weekdays to maintain this consistent revenue performance.

Section 5 — Excel vs SQL Comparison

Feature

Excel

SQL

Data Import	Manual filtering and cleaning through UI	LOAD DATA INFILE bulk-loads 185,328 rows in seconds
Analysis Method	Pivot Tables with drag-and-drop fields	SELECT, GROUP BY, SUM, COUNT queries
Visual Representation	Built-in charts and dashboards directly from Pivot Tables	No built-in visualization, requires external tools
Accuracy	Standard COUNT includes duplicate Order_IDs (over-counts orders)	COUNT(DISTINCT Order_ID) counts unique orders accurately
Scalability	Slow with large datasets, filtering 185,000 rows took 20+ minutes at times	Handles large datasets efficiently without performance issues

Ease of Use	Beginner-friendly, visual, no coding required	Requires writing queries but gives more precise control over output
Flexibility	Limited to built-in functions and UI options	Can write complex custom queries for any analysis need

Skills Demonstrated

- Excel Data Cleaning
- Duplicate Detection
- Feature Engineering
- Pivot Tables
- Pivot Charts
- Dashboard Design
- KPI Development
- Business Analysis
- SQL Data Import
- SQL Aggregations
- GROUP BY
- COUNT(DISTINCT)
- Data Validation
- Documentation

Summary:

Excel is the better tool for visual storytelling, dashboards, charts, and quick exploration of data are its strengths.

SQL is the better tool for accurate, scalable, and repeatable data analysis, especially when working with large datasets or when precision matters (like counting unique orders). Both tools produced identical business insights for this project, confirming that the analysis was correct across both approaches.